



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: Master of Computer Applications

Semester: IV

Name of the Paper: Data Science Using R

Paper Code: TMC 401

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1. (10 Marks)

- a. Define Data Science. Explain the roles of statistics, computational techniques, and domain knowledge in Data Science. (CO1)
OR
- b. Define data exploration. Discuss the key objectives of performing data exploration before applying analytical models. (CO2)

Q2. (10 Marks)

- a. What is classification? Explain its basic concept and discuss any two commonly used classification techniques. (CO1)
OR
- b. Define a dataset. Describe the various types of datasets used in the data exploration phase and their characteristics. (CO2)

Q3. (10 Marks)

- a. Describe the standard process followed for extracting meaningful insights from data. Explain the purpose of each stage involved. (CO1)
OR
- b. What are descriptive statistics? Explain how measures of central tendency and dispersion used to summarize and interpret the characteristics of a dataset during data exploration. (CO2)

Q4. (10 Marks)

- a. What is data preparation? Explain the key steps involved in preparing raw data for analysis. (CO1)
OR
- b. What is data visualization? Explain how different visualization techniques are used to represent data effectively and support understanding during data exploration. (CO2)

Q5. (10 Marks)

- a. Explain the concept of data modeling. Discuss its objectives and briefly describe the different types of models used. (CO1)
OR
- b. The following table shows the daily number of customers visiting a store over eight days:



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Day	1	2	3	4	5	6	7	8
Customers	20	18	12	9	16	25	15	13

- (a) Calculate the mean, median, and mode of the given data.
(b) Compute the range, variance, and standard deviation.

(CO2)